Lecture 33

Friday, 16 October 2026 8:23 AM

Recap: $f'(\bar{a}; \bar{y}) = \lim_{h \to 0} \frac{f(\bar{a}+h\bar{y}) - f(\bar{a})}{h}$

- Directional derivative of f at $\bar{a}$ in $\bar{y}$ direction is $\nabla f(\bar{a}) \cdot \frac{\bar{y}}{\|\bar{y}\|}$

  where $\nabla f(\bar{a}) = \left(\frac{\partial f}{\partial x}(\bar{a}), \frac{\partial f}{\partial y}(\bar{a})\right)$

- D.D. is maximum if $\bar{y} = \nabla f(\bar{a})$.

Eg1:

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0). \end{cases}$$

$$\frac{\partial f}{\partial y}(0,0) = f'((0,0);(0,1)) = \lim_{h \to 0} \frac{f((0,0)+h(0,1)) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{f(0,h) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{0 - 0}{h} = 0$$

$\frac{\partial^2 f}{\partial x^2}(0,0) = ?$

$\frac{\partial f}{\partial x}(x,y) =$ $\quad ((x,y) \neq (0,0))$

$$f(x,y) = \frac{xy}{x^2+y^2}$$

$$\frac{\partial f}{\partial x}(x,y) = \frac{(x^2+y^2)\cdot y - xy\cdot(2x)}{(x^2+y^2)^2} \qquad (x,y) \neq (0,0)$$

$$\frac{\partial f}{\partial x}(x,y) = \frac{x^2y + y^3 - 2x^2y}{(x^2+y^2)^2} = \frac{y^2 - x^2y}{(x^2+y^2)^2}$$

$\frac{\partial f}{\partial x}(0,0) = 0$ (Do it)

$$\frac{\partial^2 f}{\partial x^2}(0,0) = ?$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$

Let $g = \frac{\partial f}{\partial x}$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial g}{\partial x}$$

$$\frac{\partial^2 f}{\partial x^2}(0,0) = \frac{\partial g}{\partial x}(0,0) = \lim_{h \to 0} \frac{g(h,0) - g(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{\frac{\partial f}{\partial x}(h,0) - \frac{\partial f}{\partial x}(0,0)}{h}$$

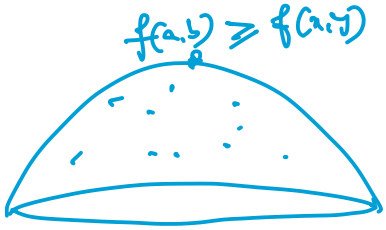
$$\frac{\partial^2 f}{\partial x^2}(0,0) = \lim_{h \to 0} \frac{0 - 0}{h} = 0$$

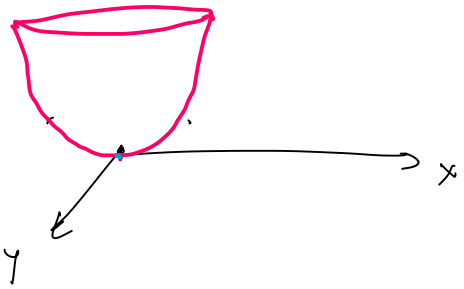
Finding Max/Minima of a multi-variable function:

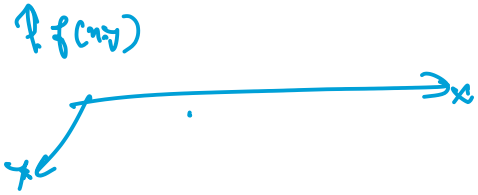
$f : \mathbb{R}^2 \longrightarrow \mathbb{R}$

Second – derivative Test

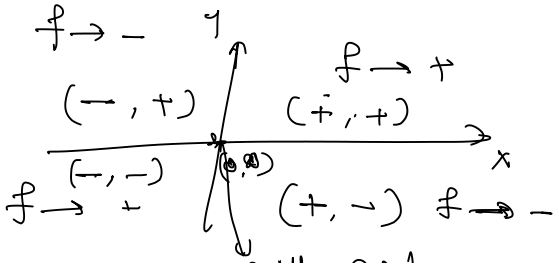
$f(x,y) = x^2 + y^2$







$f(x,y) = xy$



Saddle point.

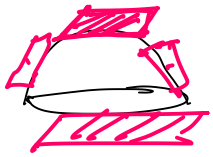
2nd derivative $(f: \mathbb{R}^2 \to \mathbb{R})$

Test: Let f be a twice diff'ble function.

(i) Find the critical points: $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$

ie find all the points where $\nabla f = \vec{0}$.

Let (P,Q) be a critical point.



(ii) Compute $\Delta = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{vmatrix}$ and $\frac{\partial^2 f}{\partial x^2}$ at (P,Q):

$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$

(a) $\Delta\big|_{(P,Q)} > 0$ & $\frac{\partial^2 f}{\partial x^2}\Big|_{(P,Q)} > 0$

then f has rel minimal at (P,Q)

(b) $\Delta\big|_{(P,Q)} > 0$ & $\frac{\partial^2 f}{\partial x^2}\Big|_{(P,Q)} < 0$

then f has rel maxima at (P,Q)

(c) $\Delta\big|_{(P,Q)} < 0$ then f has a saddle point at (P,Q)

(d) $\Delta = 0$ Inconclusive.

Eg 1): Find the extrema of $f(x,y) = xy$.

Critical points:

$\frac{\partial f}{\partial x} = 0 \longrightarrow y = 0$

$\frac{\partial f}{\partial y} = 0 \longrightarrow x = 0$

$\therefore$ $(0,0)$ is the critical point.

$\Delta = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \end{vmatrix}$ $\frac{\partial^2 f}{\partial}$

$$\begin{vmatrix} \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{vmatrix}, \qquad \frac{\partial^2}{\partial x}$$

$$\Delta = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 < 0$$

$\Rightarrow (0,0)$ is a saddle point

Eg2): $f(x,y) = x^3 + y^3 - 3xy$

Find the extrema.

$$\frac{\partial f}{\partial x} = 3x^2 - 3y$$

$$\frac{\partial f}{\partial y} = 3y^2 - 3x$$

$x^2 = y$

$y = x$

$(0,0)$, $(1,1)$, ~~$(1,-1)$~~ ✗

$(0,0)$:

$$\Delta\Big|_{(0,0)} = \begin{vmatrix} 6x & -3 \\ -3 & 6y \end{vmatrix}_{(0,0)}$$

$$\frac{\partial^2 f}{\partial x^2}\Big|_{(0,0)} =$$

$$= \begin{vmatrix} 0 & -3 \\ -3 & 0 \end{vmatrix}$$

$$= -9 < 0$$

Saddle point